

Some strange spinner... — Solution

X23 (2023) — English translation

A1

0.30 pts

Determine the moment of inertia of the system of the rod and spheres relative to the axis of rotation. Express your answer in terms of m and l .

The moment of inertia of each sphere relative to the axis of rotation is ml^2 , since their dimensions can be neglected. The original rod is equivalent to two others with mass $2m$ and length l . The moment of inertia of a rod with mass M and length L relative to the axis passing perpendicular to the rod through its end is equal to:

$$I_{\text{ST}} = \frac{ML^2}{3}$$

Then moment of inertia I of the system relative to the axis of rotation is equal to:

Answer: \textbf{Answer:}

$$I = \frac{10ml^2}{3}$$

A2

0.50 pts

Express the angular velocity ω_{i+1} of the system after the collision with the $i + 1$ emitted ball in terms of ω_i , v and l .

Let's use the laws of conservation of energy and angular momentum relative to the axis of rotation:

$$\begin{cases} E = \frac{mv^2}{2} + \frac{I\omega_i^2}{2} = \frac{mu^2}{2} + \frac{I\omega_{i+1}^2}{2} \\ L = mvl + I\omega_i = mul + I\omega_{i+1} \end{cases}$$

Here u - velocity налетающего ball/sphere and u after the collision, по направлению совпадающая со velocity v . Eliminating from системы u :

$$u = v - \frac{I(\omega_{i+1} - \omega_i)}{ml} \Rightarrow I(\omega_{i+1}^2 - \omega_i^2) = v^2 - \left(v - \frac{I(\omega_{i+1} - \omega_i)}{ml} \right)^2$$

Сокращая корень $\omega_{i+1} = \omega_i$, we get:

$$\omega_{i+1} \left(1 + \frac{I}{ml^2} \right) + \omega_i \left(1 - \frac{I}{ml^2} \right) = \frac{2v}{l}$$

Thus:

Answer: \textbf{Answer:}

$$\omega_{i+1} = \frac{7\omega_i}{13} + \frac{6v}{13l}$$

A3

0.20 pts

What is ω_1 equal to? Express your answer in terms of v and l .

Since $\omega_0 = 0$, we find:

Answer: \textbf{Answer:}

$$\omega_1 = \frac{6v}{13l}$$

A4

0.20 pts

Express ω^* in terms of v and l .

Equating ω_{i+1} and ω_i , we find:

Answer: \textbf{Answer:}

$$\omega^* = \frac{v}{l}$$

A5

0.60 pts

By solving the recurrence equation obtained in step **A1**, you will obtain an explicit expression for ω_i . Express your answer using i , v and l . Hint: Use the variable $\omega'_i = \omega_i - v/l$.

Hint: Use the variable $\omega'_i = \omega_i - v/l$.

The equation obtained in step A1 is reduced to the following form:

$$13 \left(\omega_{i+1} - \frac{v}{l} \right) = 7 \left(\omega_i - \frac{v}{l} \right)$$

Вводя переменную $\omega'_i = \omega_i - v/l$, we get:

$$13\omega'_{i+1} = 7\omega'_i$$

Hence:

$$\omega'_i = \omega'_1 \cdot \left(\frac{7}{13} \right)^{i-1}$$

Since $\omega_0 = 0$, $\omega_1 = 6v/(13l)$. Then $\omega'_1 = -7v/(13l)$, hence :

$$\omega'_i = -\frac{v}{l} \left(\frac{7}{13} \right)^i$$

and finally

Answer: \textbf{Answer:}

$$\omega_i = \frac{v}{l} \left(1 - \left(\frac{7}{13} \right)^i \right)$$

A6

0.20 pts

What would be the limiting value ω^* for such a system?

Let us note that if we put $\omega_{i+1} = \omega_i$ in the recurrent equation obtained in point A1, then for any values of the moment of inertia I the value will be:

Answer: \textbf{Answer:}

$$\omega^* = \frac{v}{l}$$

B1

1.00 pts

At what value of the parameter k_0 will the speed of the first emitted ball, immediately after colliding with the sphere, be directed horizontally?

Let u be the speed of the ejected ball immediately after impact. Since it is directed horizontally, immediately after the impact the angular momentum of the ejected ball relative to the axis of rotation is zero. Then the laws of conservation of energy and angular momentum are written as follows:

$$\begin{cases} E = \frac{mv^2}{2} = \frac{I\omega_1^2}{2} + \frac{mu^2}{2} \\ L = mvl = I\omega_1 \end{cases}$$

Let us explicitly consider the collision of an ejected ball and a sphere. Let us note that if α is the angle between the line of centers and the vertical, then $k = \sin \alpha$.

Since there is no friction, the force of interaction between them is directed along the line of centers. Then, during the impact, the velocity component of the emitted ball, perpendicular to the line of centers, remains constant and equal to $v_{\perp} = v \sin \alpha$.

Let us use the conservation of the velocity component $v_{\perp}$. Immediately after the impact, the velocity vector of the ball $\vec{u}$ is equal to:

$$\vec{u} = \vec{v} + \frac{\Delta \vec{p}_N}{m}$$

where $\Delta \vec{p}_N$ - momentum силы реакции, действующей на ball/sphere and со стороны сферы. Since $\Delta \vec{p}_N$ направлен вдоль линии центров, from рисунка we find:

$$u = v \tan \alpha$$

Substituting the value u into the system of equations from the conservation laws, we obtain:

$$v^2 = \frac{I\omega_1^2}{m} + u^2 = \frac{I}{m} \left(\frac{mvl}{I} \right)^2 + v^2 \tan^2 \alpha \Rightarrow \tan^2 \alpha = 1 - \frac{ml^2}{I}$$

Taking into account that $\sin \alpha = \frac{\tan \alpha}{\sqrt{1 + \tan^2 \alpha}}$, we find:

Answer: \textbf{Answer:}

$$k_0 = \sqrt{\frac{I - ml^2}{2I - ml^2}} = \sqrt{\frac{7}{17}}$$



